Physics 7B — Midterm 1 Review Worksheet

Berkeley (Physics 7B)

Topics for Midterm 1 review:

- Internal energy
- Thermal expansion
- Calorimetry
- Ideal gas law
- Heat transfer
- First law of thermodynamics
- Heat engines and efficiency
- Entropy and the second law

Exercises

Exercise 1. Gas mixing and partial pressures *[Ideal gas law]*

A 6.00 L container of neon gas at a pressure of 800 torr is connected to a 10.0 L container of argon gas at a pressure of 400 torr so that the gases are allowed to mix throughout both containers.

(a) Calculate the total pressure of the gas after mixing.

(b) Calculate the mole fraction of each gas after mixing.

(c) If both gases are forced together into the 10.00 L container, determine the partial pressure of each gas in the 10.00 L container.

Exercise 2. A three-process cycle *[Ideal gas law] [First law] [Entropy / Second Law]*

An ideal gas starts out in a state A, with pressure P_A, volume V_A, and temperature T_A. First, the gas expands isothermally until its volume has tripled and it reaches point B. Then, the gas cools isovolumetrically to reach point C, at which point it is adiabatically compressed to return to point A. This gas has $\gamma = \dfrac{C_P}{C_V} = \dfrac{5}{3}$. (It will be necessary to use the following relation for an adiabatic process on an ideal gas: $TV^{\gamma-1} = \text{constant}$.)

(a) Draw a P–V diagram for this cycle.

(b) What is the work done by the gas during each step of this cycle?

(c) What is the heat flowing into the gas at each step of this cycle?

(d) What is the change in entropy in the gas during each step of this cycle?

Exercise 3. Conduction through a cylinder *[Heat transfer]*

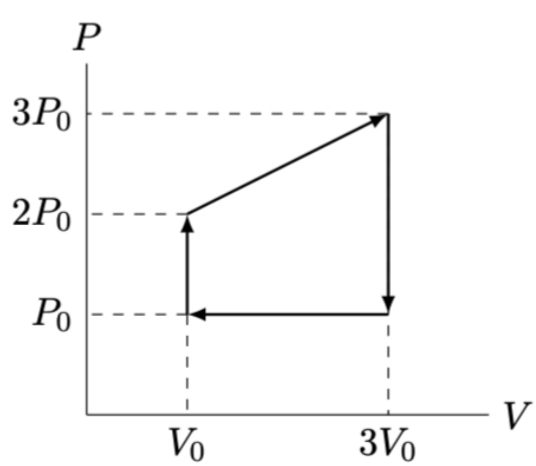
Consider the cross-sectional slice of a cylinder (a cylindrical shell) with inner radius r_1 and outer radius r_2. Assume that the cylinder has height h and is made of a homogeneous material with thermal conductivity k. The inner wall is kept at temperature T_1, and the outer wall is kept at temperature $T_2 < T_1$. The system has reached a steady state of heat transfer; i.e., the rate of heat flow is constant: $\frac{dQ}{dt} = q$.

(a) Use the conductive heat transfer equation to find an expression for the temperature gradient. Integrate to find a formula for the temperature T as a function of the radius r.

(b) Use your formula from part (a) to solve for q. (Hint: when you evaluate T at the inner and outer walls of the cylinder, what should you get?)

Exercise 4. Heat engine cycle on a P–V diagram *[Heat engines and efficiency] [First law] [Internal energy] [Ideal gas law]*

Consider the heat engine cycle shown on the following P–V diagram. Assuming that the gas is ideal and monatomic, calculate the efficiency of this engine, and compare it to the theoretical maximum efficiency.



Exercise 5. Ice–water calorimetry in an isolated container *[Calorimetry] [First law] [Internal energy]*

A container, thermodynamically isolated, contains a mass M of ice at temperature $T = 0°\mathrm{C}$. An equivalent mass M of water is now added to the container. The temperature of the water is T_W. The pressure of the water–ice system is P, the specific heat of water is C_W, and the latent heat of fusion of the ice is L. The total volume of the water and ice system does not change when the ice melts into water. Neglect the heat capacity of the container. Find:

(a) The variation of internal energy of the system of water and ice, between the initial and final state.

(b) If all the ice melts, find the final equilibrium temperature of the water and ice system.

Exercise 6. Calorimetry with container + liquid + hot pellet *[Calorimetry] [First law] [Internal energy]*

A cubic iron container with mass $M = 0.5\,\mathrm{kg}$ and side length $L = 1\,\mathrm{m}$ is half full of a strange liquid with density $\rho = 300\,\mathrm{kg\,m^{-3}}$. The system is originally at temperature $T_0 = 20°\mathrm{C}$. We then drop a very hot aluminum pellet of mass $m = 0.1\,\mathrm{kg}$ and temperature $T_{\mathrm{Al},0} = 200°\mathrm{C}$ into the liquid. The specific heats of iron, aluminum, and the liquid are $c_{\mathrm{Fe}} = 450\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $c_{\mathrm{Al}} = 900\,\mathrm{J\,kg^{-1}\,K^{-1}}$, and $c_{\mathrm{liq}} = 750\,\mathrm{J\,kg^{-1}\,K^{-1}}$, respectively. Calculate the final equilibrium temperature T_f of the system (container + liquid + pellet). Neglect any work done and assume no heat exchange with the environment.

Exercise 7. Campfire skewer + water (entropy production) *[Calorimetry] [Entropy / Second Law]*

At a campfire high in the Sierra, you are roasting marshmallows using a metal skewer of mass M_S and specific heat c_S. The skewer is at temperature T_H. You dip the skewer into a pan filled with a mass M_W of water (specific heat c_W) at temperature T_C until the skewer and the water reach thermal equilibrium (ignore the water container, and assume there is no water evaporation).

(a) What is the thermal equilibrium temperature T_F?

(b) What is the change in entropy of the skewer ΔS_S?

(c) What is the change in entropy of the water ΔS_W?

(d) What is the total change in entropy ΔS_{tot} for the system (water + skewer)?

(e) Quoting a thermodynamic law, explain and deduce the sign of ΔS_{tot}.

Exercise 8. Otto cycle *[Engines, efficiency] [First law] [Entropy / Second Law] [Ideal gas law]*

The Otto cycle comprises four segments: (1) adiabatic expansion from $A \to B$, (2) isovolumetric cooling from $B \to C$, (3) adiabatic compression from $C \to D$, and (4) isovolumetric heating from $D \to A$.

(a) Sketch the P–V diagram.

(b) What is the heat absorbed in each segment in terms of P and V?

(c) Calculate the efficiency $e = W/Q_H$ of this engine for an ideal diatomic gas, where W is the work done *by* the gas in a full cycle and Q_H is the total heat flowing into the gas. Compare this to the efficiency of the Carnot cycle.

(d) Would the efficiency increase or decrease for a monoatomic gas? Show the comparison explicitly.

(e) Calculate the change in entropy across the whole cycle.